

Experiment-4-Intermediate-2

Title: FPGA-Based Event-Driven Spiking Neural Network (SNN) Accelerator for Handwritten Digit Recognition

Objective:

The objective of this experiment is to design and implement a hardware-oriented Spiking Neural Network (SNN) on an FPGA for handwritten digit classification using the MNIST dataset. The system converts image data into sparse spike sequences using delta encoding, processes the spikes through a fully connected synaptic layer with fixed-point weights, and generates neuron outputs using a Leaky Integrate-and-Fire (LIF) neuron model with shift-based leakage.

The complete SNN is described using Verilog HDL, validated through ModelSim RTL simulation, and targeted for implementation on the Terasic DE10-Nano FPGA board. The design explores the use of sparsity, fixed-point arithmetic, and hardware-friendly neuron computation to make SNN processing suitable for digital FPGA hardware.

Block Diagram:

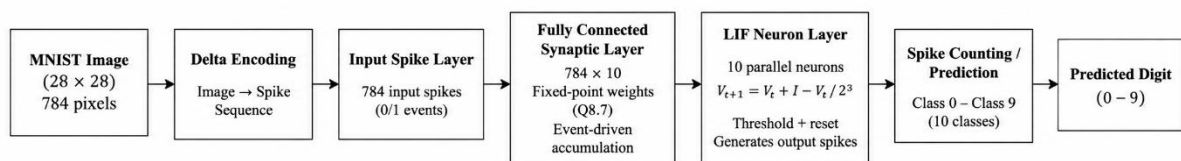


Fig. 1. Overall system architecture of the FPGA-based SNN for MNIST digit classification.

The system accepts a 28×28 MNIST handwritten digit image containing 784 pixels. The image is converted into a temporal sequence of sparse spikes using delta encoding. The resulting spike sequence is supplied to the fully connected synaptic layer, where the active input spikes are multiplied by their corresponding fixed-point weights and accumulated to generate synaptic currents for ten output neurons. Each current is processed by a Leaky Integrate-and-Fire (LIF) neuron. The LIF neurons use shift-based leakage and generate output spikes when their membrane potential reaches the threshold. The output spike activity is used to determine the predicted digit class from 0 to 9.

RTL Design:

The SNN implementation is developed using a combination of Python-based preprocessing and Verilog HDL-based hardware modules. The delta encoding stage is implemented in Python and converts the MNIST images into sparse temporal spike sequences. These generated spike sequences are then sent to the Verilog-based SNN for hardware-oriented processing. The RTL implementation consists of the `snn_top.v`, `fc_layer.v`, and `lif_neuron.v` modules, along with dedicated testbenches `tb_snn_top.v` and `tb_fc_layer.v` for functional verification.

1. `fc_layer.v`

The `fc_layer` module implements the fully connected synaptic layer of the SNN. It accepts a 784-bit input spike vector, where each bit represents the presence or absence of a spike from one MNIST pixel,

and produces ten signed synaptic currents, one for each output digit class. The module stores 7,840 trained weights corresponding to the 784×10 connections. These weights are loaded from the exported memory file and represented using Q8.7 fixed-point format. When an input spike is active, the corresponding weight is accumulated into the respective output current. Thus, the module performs the weighted accumulation required to transform the sparse input spike pattern into ten synaptic currents.

2. lif_neuron.v

The `lif_neuron` module implements the Leaky Integrate-and-Fire neuron model used for spike generation. Each neuron receives a signed synaptic current from the fully connected layer and maintains an internal membrane potential. At every enabled clock cycle, the membrane potential is updated by adding the incoming current and subtracting a leakage term. The leakage is implemented using an arithmetic right-shift operation with `LEAK_SHIFT = 3`, corresponding to division by 2^3 , which provides a hardware-friendly implementation. When the updated membrane potential reaches or exceeds the specified threshold, the neuron generates a one-cycle output spike and resets its membrane potential to zero. Ten instances of this module are used in the complete SNN to represent the ten output classes.

3. snn_top.v

The `snn_top` module is the top-level integration module of the hardware SNN. It connects the `fc_layer` with the LIF neuron layer and controls the flow of the input spike data through the network. The 784-bit input spike vector is provided to the fully connected layer, which generates ten synaptic currents. These currents are then supplied to ten LIF neuron instances, producing ten output spike signals corresponding to digit classes 0–9. The module also provides the clock, reset, enable, and completion signals required for coordinated operation of the SNN.

4. Testbench and Verification

Two testbenches were used for RTL verification. The `tb_fc_layer.v` testbench verifies the functionality of the fully connected layer by providing input spike patterns and observing the generated synaptic currents. The `tb_snn_top.v` testbench verifies the complete SNN by supplying the encoded MNIST spike sequences, processing multiple time steps, and observing the output spikes and predicted digit classes. The complete SNN testbench was used for the final ModelSim evaluation, where 91 out of 100 MNIST test images were classified correctly, giving an RTL simulation accuracy of 91%.

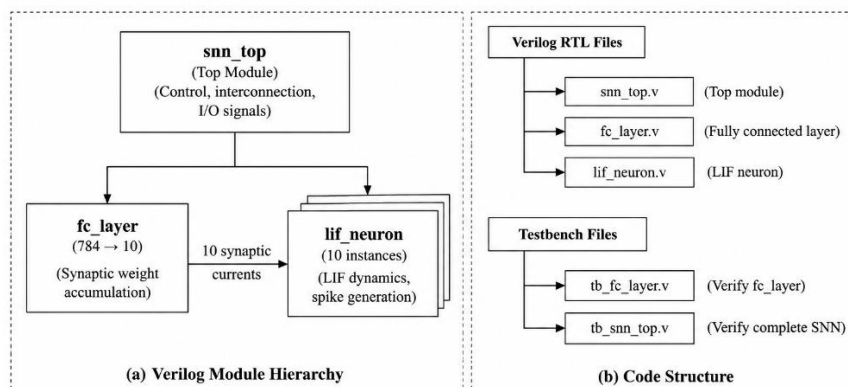
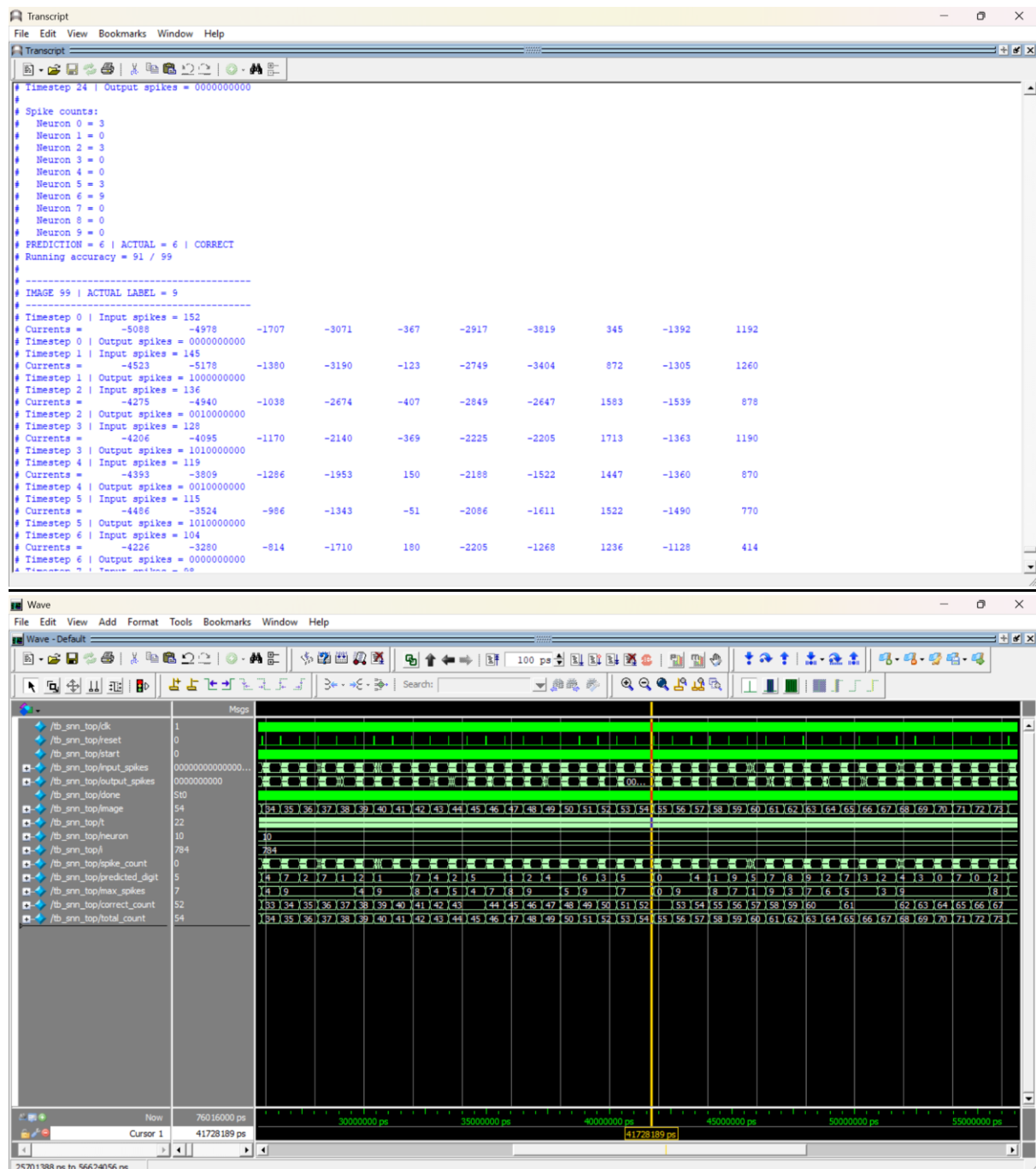


Fig. 2. Verilog module hierarchy and code structure of the FPGA-based SNN.

Simulation Results:

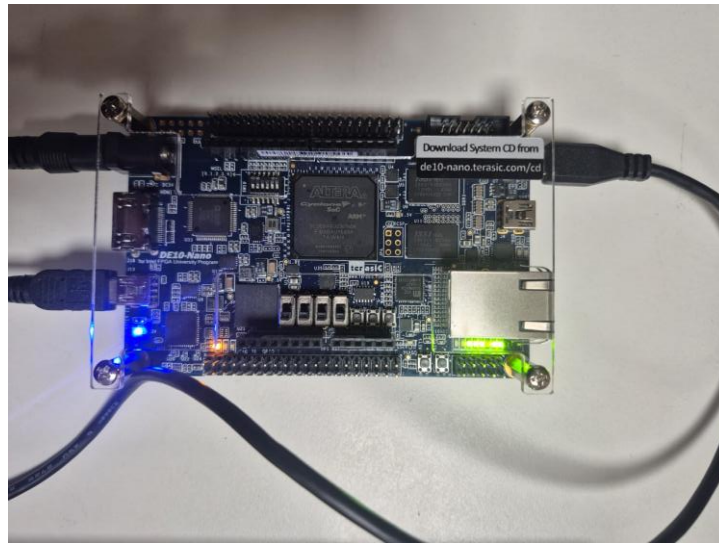


```
=====
100-IMAGE TEST COMPLETE
=====

Correct predictions = 91
Total images       = 100
Accuracy           = 91%
=====

** Note: $finish      : C:/intelFPGA/18.1/tb_snn_top.v(383)
Time: 76016 ns  Iteration: 0  Instance: /tb_snn_top
1
```

Hardware Implementation:



Applications:

The FPGA-based SNN developed in this experiment demonstrates the use of spiking neural computation for image classification on resource-constrained digital hardware. The architecture can be adapted for applications that require event-based processing and low-complexity neural computation, such as handwritten character recognition, edge-based image classification, embedded pattern recognition, and intelligent sensor systems.

The use of sparse spike encoding and hardware-friendly LIF neurons makes the architecture relevant to edge-AI and embedded neuromorphic systems, where neural processing may need to be performed locally rather than relying entirely on a CPU or cloud platform. The same architecture can also serve as a foundation for extending SNN-based processing to other classification tasks and event-driven datasets.

Conclusion:

The experiment successfully demonstrated an FPGA-oriented SNN for MNIST digit classification using sparse spike encoding, fixed-point synaptic processing, and LIF neurons. The RTL design was verified in ModelSim with 91% accuracy on 100 test images. The work provided practical experience in SNNs, Verilog RTL design, fixed-point computation, and FPGA-based neural processing.